

Square Wave Generation - Hard Problem

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You are designing a system to drive something known as a capacitive discharge igniter (CDI). These are used to drive spark plugs, and have a relatively simple interface. There are three input wires; two are for power and ground, and one is signal. When signal is high, the CDI is charging. When signal goes low, it fires a spark in the attached spark plug.

Design a circuit that outputs a 100hz square wave that is high for 9.5ms and low for 0.5ms. You are welcome to use an IC like a 555 timer. Bonus points if you do it with only resistors, capacitors, inductors, diodes, and an op-amp.